MTH357

Timed Online Assignment – January Semester 2021

Applied Regression Analysis I

Thursday, 20 May 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **NINE (9)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

The quality of red wine is thought to be related to the properties of clarity, aroma, body, flavour and oakiness. 38 red wines were evaluated by experts and data summary is shown in Table Q1-I and Figure Q1 below:

Index No	Clarity	Aroma	Body	Flavour	Oakiness	Quality	Region
1	0.5	3.3	5.4	4.3	3.6	12.3	2
2	0.7	7.7	6.6	6.7	3.7	16.1	3
3	0.8	5.9	5.7	7	4.1	16.1	3
…	…	…	…	…	…	…	…
38	0.9	3.4	5	3.4	3.4	7.9	2

Table Q1 - I

```
   Clarity           Aroma            Body          Flavour
Min.   :0.5000   Min.   :3.300   Min.   :2.600   Min.   :2.900
1st Qu.:0.8250   1st Qu.:4.125   1st Qu.:4.150   1st Qu.:4.225
Median :1.0000   Median :4.650   Median :4.750   Median :4.800
Mean   :0.9237   Mean   :4.847   Mean   :4.684   Mean   :4.768
3rd Qu.:1.0000   3rd Qu.:5.450   3rd Qu.:5.375   3rd Qu.:5.500
Max.   :1.0000   Max.   :7.700   Max.   :6.600   Max.   :7.000
   Oakiness        Quality          Region
Min.   :2.900   Min.   : 7.90   Min.   :1.000
1st Qu.:3.700   1st Qu.:11.15   1st Qu.:1.000
Median :4.100   Median :12.45   Median :2.000
Mean   :4.255   Mean   :12.44   Mean   :1.868
3rd Qu.:4.775   3rd Qu.:13.75   3rd Qu.:3.000
Max.   :6.000   Max.   :16.10   Max.   :3.000
```

Figure Q1

The summary of the regression results from R is shown below in Table Q1-II:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	4.00	2.23		0.08
Clarity	2.34	1.73		0.19
Aroma	0.48	0.27		0.09
Body	0.27	0.33		0.42
Flavour	1.17	0.30		0.00
Oakiness	-0.68	0.27		0.02

Table Q1 – II

R^2: 0.7206
$R^2 - adj$: 0.6769
F: 16.51 on 5,32 $DF, p - value <$ 0.0001

(a) The data is stored in the folder "D:/SUSS2021/MTH357" under "Wine Data.csv". Write down the R code which will read the data from the file, generate the summary statistics and fit the multiple linear regression model above. Note that the model above does not include "Region" as an independent variable.

(8 marks)

(b) Write down the fitted regression model. Interpret the coefficients for Flavour and Oakiness, and discuss the limitations to the model.

(7 marks)

(c) Compute the missing t-values in Table Q1-II, and perform t-tests to assess the contribution of each regressor to the model. What conclusions can you draw?

(8 marks)

(d) The analyst decided to exclude the variable "Body" from the model. (i.e. the new model has Clarity, Aroma, Flavour and Oakiness as X variables). What will happen to the R^2 and $adj\ R^2$? Write down a possible value for the new R^2 and $adj\ R^2$, and explain your answer clearly.

(8 marks)

(e) There are a number of variables which are not statistically significant under the individual t-tests. If we were to use nested model F-test instead to evaluate if the variable is worth including in the model, will we always reach the same conclusion as the t-test?

(3 marks)

For single variable

→ compare 1 by 1

Question 2

(a) Explain Cook's Distance and how can it be used to assess the model in regression analysis.

(4 marks)

(b) The Cook's Distance from the Regression in Question 1 was computed and visualized in the Figure Q2(b) below. The dependent variable is Quality, and while the independent variables are clarity, aroma, body, flavour and oakiness. Find the appropriate F value and hence evaluate the Cook's Distance.

90th percentile

0.91

maybe?

definitely

no is usually close to 1

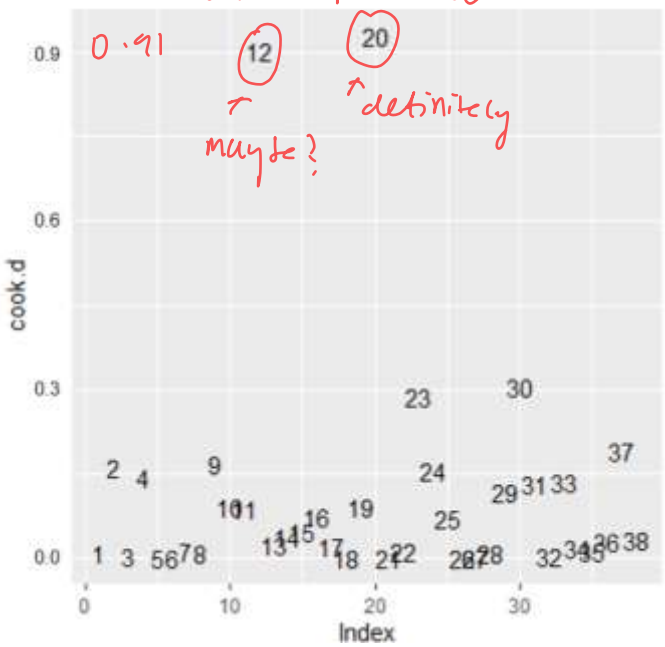

Figure Q2(b)

(5 marks)

(c) One of your colleagues felt that the points 12 and 20 in Figure Q2(b) above should be excluded from the analysis. Is it best practice to exclude data which might be outliers? Discuss.

(3 marks)

No? ∵ not data fault. is your model's fault.

(d) In simple linear regression, the standard error of the OLS estimate for slope is given by:

× variance

→ variance

$$\sigma^2_{\hat{\beta}_1} = \frac{\sigma^2}{S_{xx}}$$

represents uncertainty in your data

where σ^2 may be estimated by

cannot be reduced to 0

$$s^2 = \frac{SSE}{n-2}.$$

→ ∵ every new data pt, SSE ↑ : when ↑n → ↑SSE

means more data pt

$S_{xx} \to \infty$

$\therefore \sigma^2_{\hat{\beta}_1} \to 0$

$s^2 \not\to 0$

By considering both numerator and denominator, discuss what happens if the sample size is increased by a large amount (towards infinity). Can the standard error of the slope be reduced infinitesimally close to 0 with a greater sample size?

(4 marks)

Question 3

qlty = $\beta_0 + \beta_1$ flavour + $\beta_{21} R_1 + \beta_{22} R_2$, set R_3 as baseline

is R_3 wine better than R_1 & 2?

The Wine Appellation system is a method where countries categorize their wines into specific grape-growing **geographical regions.** If the wine production meet certain stringent conditions, the wines can be associated with an appellation - a stamp of "high quality". For instance, since 1937, France qualify their wines with the Appellation d'Origine Contrôlée (AOC) system. Famous appellations in France include the Champagne & Bordeaux, amongst hundreds of others.

We would like to know if Wine Quality is affected by Region. In the data from Question 1, the wines come from 3 regions, tagged as Region 1, Region 2 and Region 3. Region 3 are the geographical regions with appellations.

(a) Using Regression Analysis, demonstrate how we can investigate if Quality is affected by Region.

For simplicity, assume that we only have Quality (y), Flavour (x_1) and Region (x_2) in our dataset. Flavour is known to affect Quality and must be included in the Regression Model.

Your answer must include (marks in parenthesis):

compare region 3 and not region 3

(i) The Regression Model that you propose to use.

(4 marks)

(ii) The hypotheses which are to be validated.

(3 marks)

set region 3 as the baseline (recommendation)

(iii) Even without running the regression, you should already have an expectation of how the coefficients should be. Write down the expected signs (positive or negative) and magnitude of the coefficients.

(3 marks)

if high y = ↓tr qlty then expect $\hat{\beta}_{21}$ & $\hat{\beta}_{22}$ to be −ve

(∵ region 3 supposed to be better)

qlty is 7 to 16
so if (-2, -3)
is like
-10, means
sth is
wrong.

(-2, -3) ← So R_1 & R_2 should cause a ↓ in y value (↓ in qlty)

(iv) The statistical tests to be conducted, stating the distribution clearly.

→ nested F test, $\beta_{21} = \beta_{22} = 0$

(4 marks)

(v) How the results may be interpreted?

(2 marks)

Marks will be given for detailed but concise answers. Make sure to explain your reasoning clearly.

(b) Explain the need to include Flavour into the model if we know for a fact that it affects Quality.

→ baseline difference between preference.

→ compensate

(4 marks)

(c) Using the same wine quality data, we fit a model that relates wine Quality (y) to Flavour (x_1) and Region (x_2), but this time, we use Region as an allocated code, taking on the values shown in the data (1, 2 and 3s), instead of treating it as a categorical variable. Discuss the interpretation of the model in this model. Under what situation will this be a reasonable approach?

?

(5 marks)

Question 4

In a study conducted in 2017-2018, a research team from the Lee Kuan Yew School of Public Policy estimated that Singapore seniors each need at least $1,379 a month to meet basic needs. This can be fulfilled by a combination of savings and investment.

To estimate if the population is saving sufficiently to meet retirement needs, a researcher interviewed 15 working people and asked for their savings. He wants to analyse the relationship between age (x) and savings (y) by using regression analysis.

$y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim N(0, \sigma^2)$

Here are his findings as seen in Figure Q4 below:

	Age	Savings / month (SGD)
1	46.6	1012
2	43	991
3	35.4	995
4	31.6	958
5	29.3	864
6	28.7	865
7	28	964
8	27.7	852
9	25.1	801
10	23.8	658
11	23.8	700
12	23.7	721
13	23.7	747
14	21.3	653
15	18.9	722

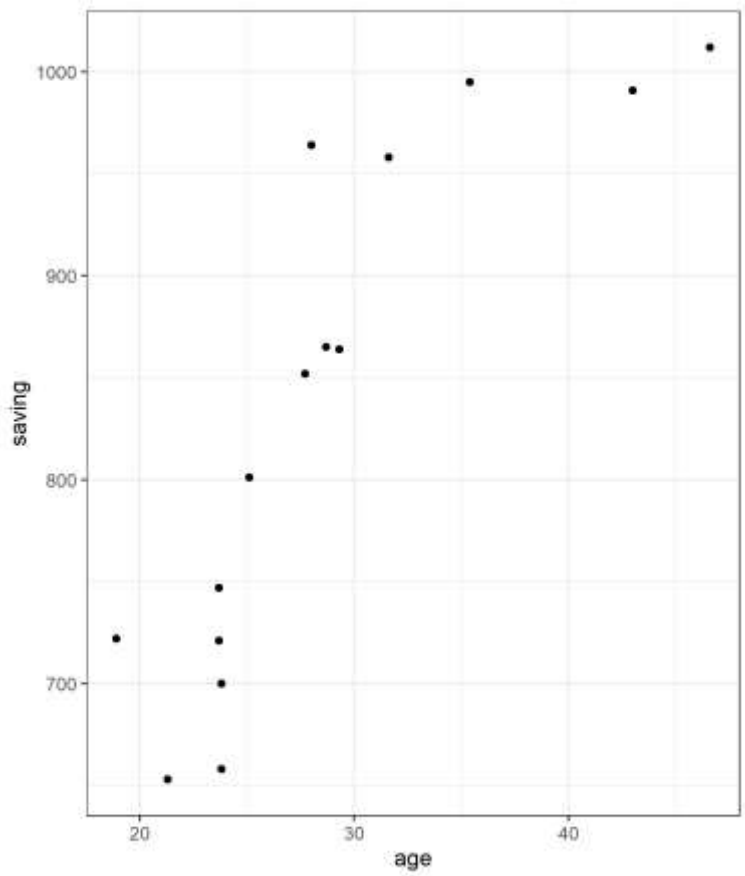

Figure Q4

$\rightarrow \bar{x}, \bar{y}$

(a) Calculate the summary statistics and hence compute the coefficients for a simple linear regression model.

s^2 / S_{xx}

(6 marks)

(b) Given that $s = 70.97$ on 13 df, perform a statistical test to verify if β_1 is statistically significant. What assumptions have you made in the statistical test?

y linear, errors ~ N, independent etc... (standard stuff)

(5 marks)

(c) Using the linear regression model obtained in Question 4(a), analyse the difference in the expected savings per month of a 45 year old versus a 25 year old? Construct a 95% confidence interval for the expected difference. sub in

(6 marks)

(d) Using the chart given in the Question or otherwise, draw the (OLS) simple regression line. Comment on the fit.

(4 marks)

(e) The research supervisor suggested to use polynomial terms to improve the model fit. Discuss how can we implement this, the sign of the coefficient (positive or negative) and how can we validate if the new model is superior to the current model?

(4 marks)

----- END OF PAPER -----

4c) sub in & substract.

CI: need to have & variance of the difference

$Var(\hat{y}_{45} - \hat{y}_{25})$

$= Var(20\hat{\beta}_1) = 400\, Var(\hat{\beta}_1) =$ _____

SD = $\sqrt{\quad}$

$t_{0.975,\,13} =$ _____

(a) Calculate the summary statistics and hence compute the coefficients for a simple linear regression model.

(6 marks)

(b) Given that $s = 70.97$ on 13 df, perform a statistical test to verify if β_1 is statistically significant. What assumptions have you made in the statistical test?

(5 marks)

(c) Using the linear regression model obtained in Question 4(a), analyse the difference in the expected savings per month of a 45 year old versus a 25 year old? Construct a 95% confidence interval for the expected difference.

(6 marks)

(d) Using the chart given in the Question or otherwise, draw the (OLS) simple regression line. Comment on the fit.

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(4 marks)

----- END OF PAPER -----

e) Sug:

Scatterplot looks curvy-

① model should be sup when use F/t test

② cross validate